GRADED ASSESSMENT · TAKE-HOME

Pandas: Ingestion, Cleaning & Slicing

Total: 100 marks · Suggested time: 2–3 hours · Open resources

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INSTRUCTIONS: READ BEFORE STARTING

You may use any reference, including AI tools. **But note:** the marks live in your **reasoning**, not in code that runs. Several questions have *no single correct fix*. You are graded on whether your justification names the trade-off you accepted and the one you rejected.

Every code answer must be accompanied by a **2–4 sentence written rationale**. A correct output with no reasoning earns roughly half marks. A wrong output with sharp reasoning often earns more.

Where a question asks you to *predict before running*, write the prediction **first**, then the actual result, then explain any gap. Answers that show only the final result lose the prediction marks.

All questions refer to `saas.csv`, loaded as strings so nothing is silently coerced:

```
import pandas as pd
df = pd.read_csv('saas.csv', dtype=str)
```

The file has columns: `account_id, company, signup_ts, country, seats, mrr, plan, last_active, churned`. A preview is on the last page for reference.

A Section A: The Mental Model

These test whether you can reason about *what pandas is doing*, not just produce output.

Q1. Predict the alignment [10 marks]

Without running anything, predict the exact output of the code below, including every index label and every value, including anything that is not a number. Then run it and reconcile.

```
a = pd.Series({'jan': 100, 'feb': 200, 'mar': 300})
b = pd.Series({'feb': 10, 'mar': 20, 'apr': 40})
result = a + b
```

In 3–4 sentences, explain *why* the result has the number of rows it does, and why some cells hold the value they hold. Name the mechanism responsible.

Submit: Your written prediction, the actual output, and the explanation. **Do not** skip straight to the run.

Q2. Series or DataFrame? [8 marks]

For each expression below, state whether it returns a Series or a DataFrame, and give one concrete reason the difference would matter in later code (name a specific operation that behaves differently).

```
df['mrr']
df[['mrr']]
df.loc[:, 'mrr']
df.loc[:, ['mrr', 'seats']]
```

Submit: A short table: expression, returned type, and the consequence you identified.

B Section B: Column Slicing

Q3. Four routes to the same columns [12 marks]

Write **four different** one-line expressions that each select *exactly* the columns `company`, `seats`, and `mrr`, no more and no fewer. Use a genuinely different mechanism each time (by name, by `.loc` label range, by `.iloc` position, and by `.filter` or `select_dtypes`).

Then answer: two of your four routes are *fragile* if a colleague reorders or renames columns upstream. Which two, and why? Which single route would you put in production code, and why?

> **Submit:** The four expressions, plus your fragility analysis and production choice with justification.

Q4. The inclusive/exclusive trap [8 marks]

Suppose the columns are in the order listed above. Predict, before running, how many columns each of these returns and name them:

```
df.loc[:, 'company':'mrr']
df.iloc[:, 1:5]
```

They look like they should return the same thing. Do they? Explain the rule that governs each, in your own words.

> **Submit:** Both predictions, the actual column lists, and the rule explanation.

C Section C: Diagnosis and Cleaning

This section is the core of the assessment. There are **deliberate traps** where the obvious fix is wrong. Marks are awarded for catching them, not for clean-looking code.

Q5. The count is the question [12 marks]

Open the data and list every data-quality problem you can find. For each, state (a) what the problem is, (b) which of the four questions it belongs to (*what is one row / what should each column be / what is missing or lying / what shape do I need*), and (c) whether you would fix it, and why.

At least one "problem" in this dataset is **not actually dirt**. Identify it and defend leaving it alone. Full marks require you to distinguish real dirt from legitimate data.

> **Submit:** A diagnosis table with your four-questions classification and fix/no-fix decision per item.

Q6. The duplicate that isn't [12 marks]

Two account IDs appear more than once. A classmate proposes a single line: `df = df.drop_duplicates()`. Predict exactly what this line does to *each* repeated account, then run it and check.

Explain why `drop_duplicates()` is the wrong tool for at least one of the repeats. Then write the code you would actually use, and state the one piece of **business information** you would need before committing to it.

> **Submit:** Prediction, actual result, the flaw you found, your replacement code, and the missing information you named.

Q7. One column, several diseases [14 marks]

Compute the implied revenue per seat (`mrr` divided by `seats`, as numbers). Most accounts cluster around one value. Using that baseline, at least four rows look wrong **for different reasons**.

For each suspicious row, give your single *most likely* explanation and the fix it implies. Crucially: for at least one row, there are **two** competing explanations the data alone cannot distinguish. Identify that row and explain what extra evidence would settle it.

> **GRADING NOTE: READ THIS**
> A generic answer like "remove outliers with the IQR method" earns **almost nothing** here. Outliers in this column are not a single phenomenon. Your marks come from showing the *different causes* behind numerically similar anomalies.
>
> **Submit:** Your per-seat computation, a per-row explanation with fixes, and the row with two irreconcilable readings.

Q8. Parsing is not validating [10 marks]

The date columns contain more than one kind of problem. Some can be caught by parsing; at least two **cannot**, because the values are valid dates that are nonetheless impossible.

Write code that surfaces *both* kinds. Then explain, in your own words, the difference between a **parsing** problem and a **validation** problem, using a specific example of each from this dataset.

> **Submit:** Code that flags both kinds, plus your parsing-vs-validation explanation with named examples.

D Section D: Judgement

Q9. The same null, two fates [8 marks]

One row is missing a value in a *label* column; another is missing a value in a *measure* column. Identify both.

Argue why the **same** action (say, dropping the row) could be correct for one and wrong for the other. Tie your argument to a specific downstream use of the data.

> **Submit:** Both rows identified, and your argument tied to a named downstream use.

Q10. Cleaning is a function of the question [6 marks]

Pick any two cleaning decisions you made earlier in this assessment. For each, describe a realistic goal under which your decision would **reverse**, you would do the opposite.

Conclude in two sentences: what does this tell you about whether "clean" is a property of a dataset?

> **Submit:** Two reversible decisions with their triggering goals, and your two-sentence conclusion.

How This Is Marked

Each question is scored across three bands. Aim for the top band by making your **reasoning visible**.

Top band	Correct or well-defended answer WITH reasoning that names the mechanism or trade-off. Catches the trap. Distinguishes real dirt from legitimate data.

Middle band	Reasonable answer but reasoning is thin, generic, or borrowed. Runs correct code without explaining why. Misses the subtler trap but handles the obvious problems.

Low band	Output only, no rationale. Applies a generic recipe (drop all nulls, remove outliers) that ignores the domain. Confident but wrong, with no sign of having questioned the data.

WHY REASONING IS WEIGHTED THIS HEAVILY

A tool can produce plausible pandas code for any of these prompts. What it cannot reliably do is know that `A-1002`'s repeat is a real update, that `-99` is a code rather than a price, or that a valid future date is impossible in context. **Those judgements are what this assessment measures.** Show them, and the marks follow.

Appendix: Data Preview

The full file is provided separately. This preview is for reference while reading the questions.

```
account_id,company,signup_ts,country,seats,mrr,plan,last_active,churned
A-1001,Acme Corp,2023-01-05 09:00:00,US,25,2500,enterprise,2024-03-01,False
A-1002,Beta LLC,2023/02/14,US,5,450,pro,2024-03-02,False
A-1001,Acme Corp,2023-01-05 09:00:00,US,25,2500,enterprise,2024-03-01,0
A-1003,Cirrus,1675209600,GB,0,0,pro,2024-02-28,False
A-1004,Delta Inc,2023-03-20,United States,12,1188,pro,2024-03-01,No
A-1005,Echo,2023-04-01,US,8,-99,pro,2024-03-03,False
A-1006,Foxtrot,2023-05-11,us,3,270,PRO,2024-03-02,FALSE
A-1007,Gamma,2023-06-30,DE,15,1800,enterprise,2023-06-15,True
A-1008,Helio,2023-07-22,US,,540,pro,2024-03-04,False
A-1009,Iris,2023-08-08,US,10,900,pro,,True
A-1010,Juno,2023-09-01,US,20,0,enterprise,2024-03-01,False
A-1011,Kilo,2023-10-10,CA,7,630,pro,2024-03-05,false
A-1002,Beta LLC,2023-02-14,US,6,540,pro,2024-03-06,False
A-1012,Lima,2023-11-11,US,1000000,90,pro,2024-03-02,False
A-1013,Mike,2023-12-25,US,4,360,pro,2099-01-01,False
A-1014,,2024-01-15,US,9,810,pro,2024-03-03,False
A-1015,Oscar,2024-02-01,US,6,540,starter,2024-03-04,False
```

Submission link :
https://docs.google.com/forms/d/e/1FAIpQLSeyb2di_jJtWwExtzBBOg27QnDkq_azrL8U4HYWrcQIJDHUvw/viewform?usp=send_form